

TIME SERIES FORECASTING

Business Report

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Problem Statement

Context:

In the dynamic landscape of the financial markets, the demand for accurate predictions of gold prices is crucial for investors, traders, and stakeholders. Similar to the challenges faced by business communities in the United States in identifying and attracting the right talent, predicting the future values of precious metals poses a significant challenge. The inherent volatility and complex dynamics influencing gold prices require a reliable forecasting tool.

The goal is to provide a valuable asset for individuals navigating the financial markets, ensuring they can approach the unpredictability of gold prices with confidence and strategic insight.

Objective:

In the realm of financial markets, predicting the future value of gold and silver holds significant importance for investors, traders, and stakeholders. The challenge lies in the inherent volatility and complex dynamics influencing gold prices. To address this, the objective is to develop an accurate time series forecasting model that can predict the future prices of gold. The model should leverage historical gold and silver price data. The anticipated outcome is a robust forecasting tool that empowers stakeholders to make informed decisions and navigate the dynamic landscape of the gold market with confidence.

1 - Define the problem and perform Exploratory Data Analysis:

-Data Overview

	Date	Price
0	08/17/2023	1915.2
1	08/16/2023	1928.3
2	08/15/2023	1935.2
3	08/14/2023	1944.0
4	08/11/2023	NaN

Table 1- Top 5 rows of the dataset

	Date	Price
2534	08/23/2013	NaN
2535	08/22/2013	1370.8
2536	08/21/2013	1370.1
2537	08/20/2013	1372.6
2538	08/19/2013	1365.7

Table 2- Last 5 rows of the dataset

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2539 entries, 0 to 2538
Data columns (total 2 columns):
#   Column  Non-Null Count  Dtype  
---  -
0   Date    2539 non-null      object  
1   Price   2460 non-null      float64
dtypes: float64(1), object(1)
memory usage: 39.8+ KB
```

Table 3- Basic info of the dataset

	count	mean	std	min	25%	50%	75%	max
Price	2460.0	1468.78435	282.977384	1049.6	1244.375	1322.15	1775.025	2069.4

Table 4- Statistical Summary of the Dataset

```
Date      0
Price     79
dtype: int64
```

>Observation:

- The dataset has 2539 rows and 2 columns.
- The dataset has 2 datatypes date are object datatype and price are float64 datatype.
- Price column has 79 missing values, so we need to treat the missing value.
- Mean of the price is 1468.78 and std is 282.98 and max value is 2069.4.

-Missing value treatment

```
Date      0
Price     0
dtype: int64
```

>Observation:

- We are treating the missing value of the Price by its mean.
- The Date column is no more needed for further analysis, so we dropped the Date column.

Exploratory Data Analysis;

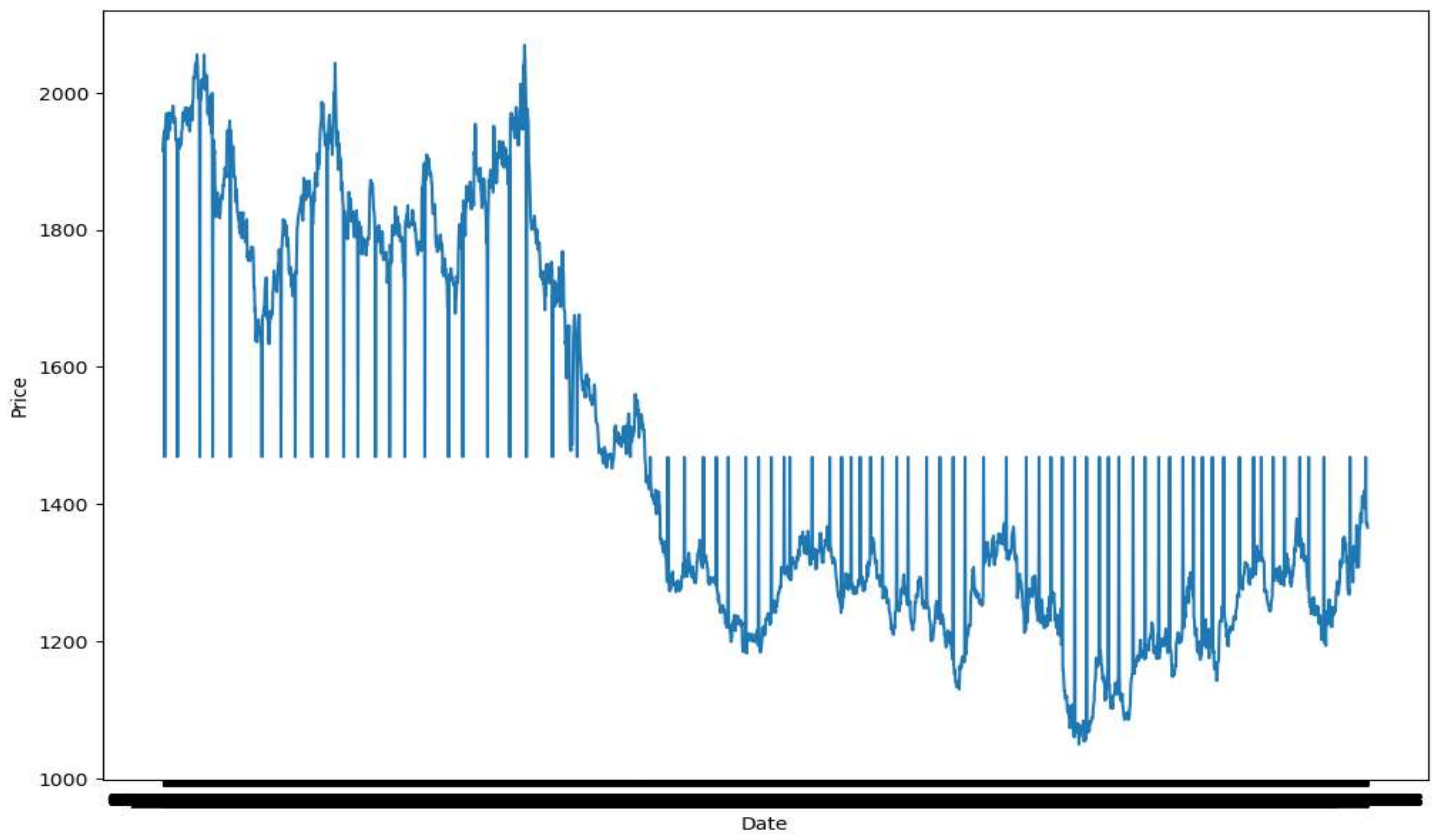


Figure 1

#We will resample the data to get values at a month-level from the day-level data by taking the average over the months.

Price	
Date	
2013-08-31	1398.908435
2013-09-30	1348.461905
2013-10-31	1324.064537
2013-11-30	1273.433333
2013-12-31	1235.049289

Table 5- Resampling the data

	count	mean	std	min	25%	50%	75%	max
Price	121.0	1472.077843	274.950834	1086.686276	1254.056398	1330.680952	1765.860189	1997.940909

Table 6- Statistical summary after resampling

>observations: After resampling the data have present 121 rows and 1 column, and mean of the Price 1472.07, std of Price is 274.95 and max value of Price is 1997.94.

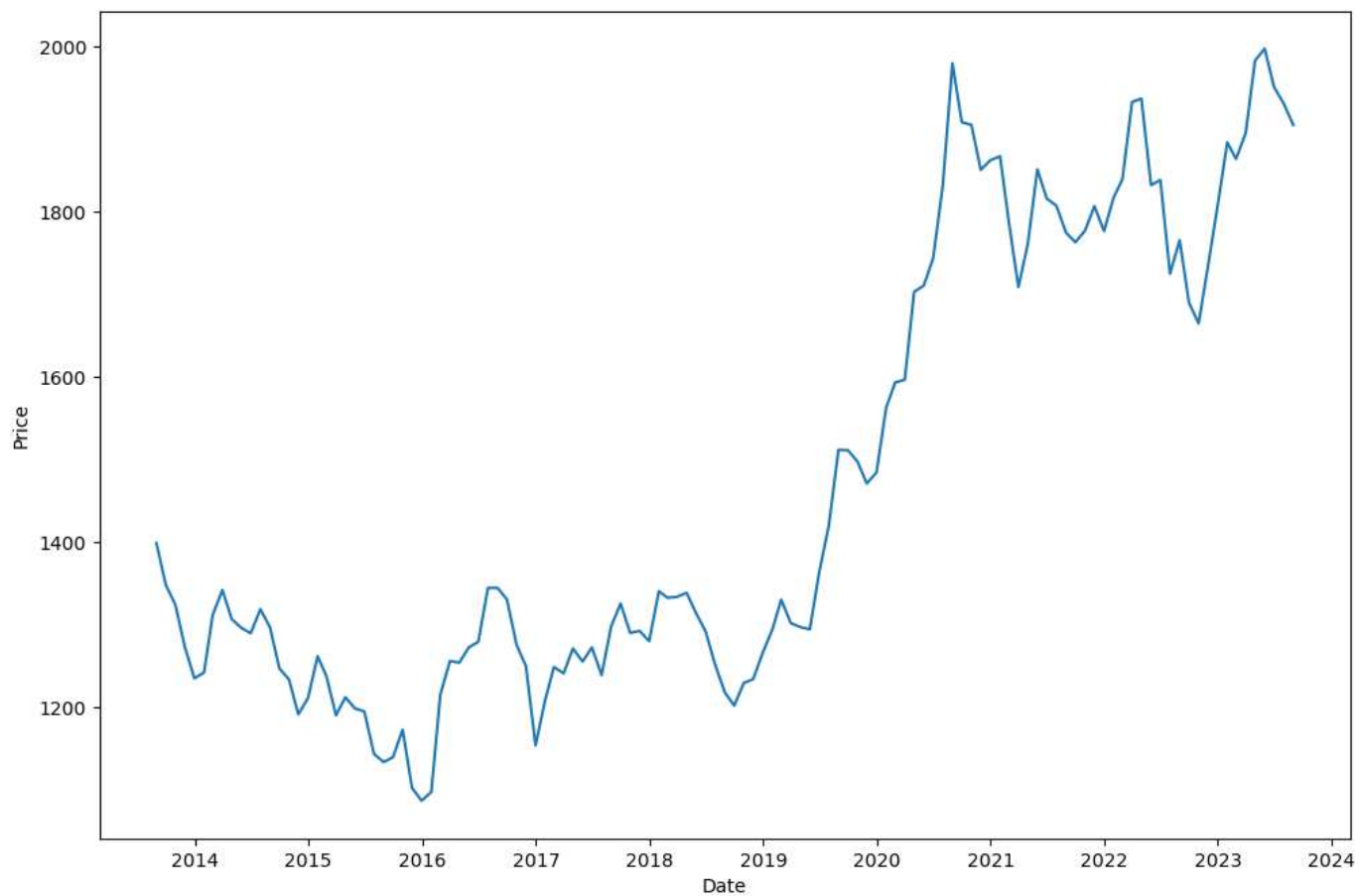


Figure 2

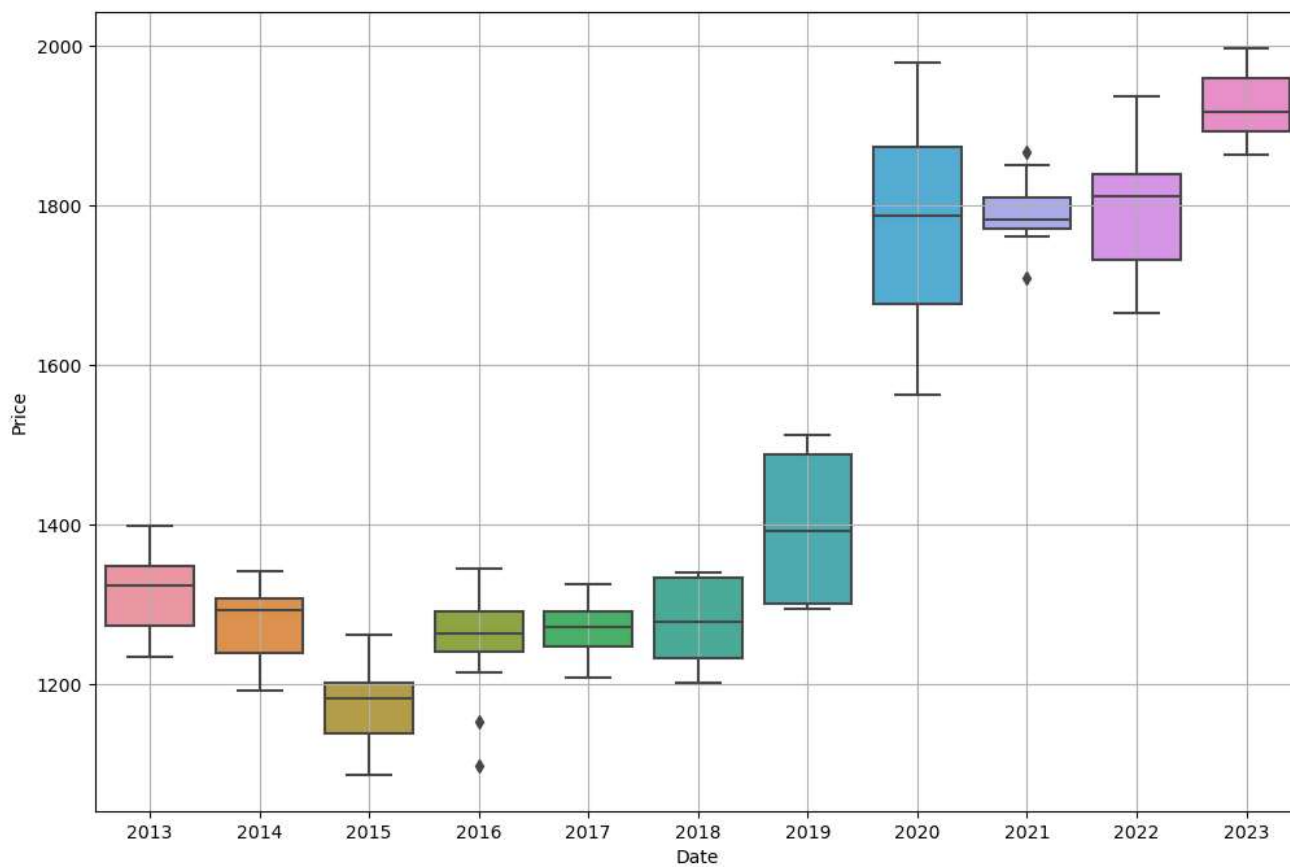


Figure 3

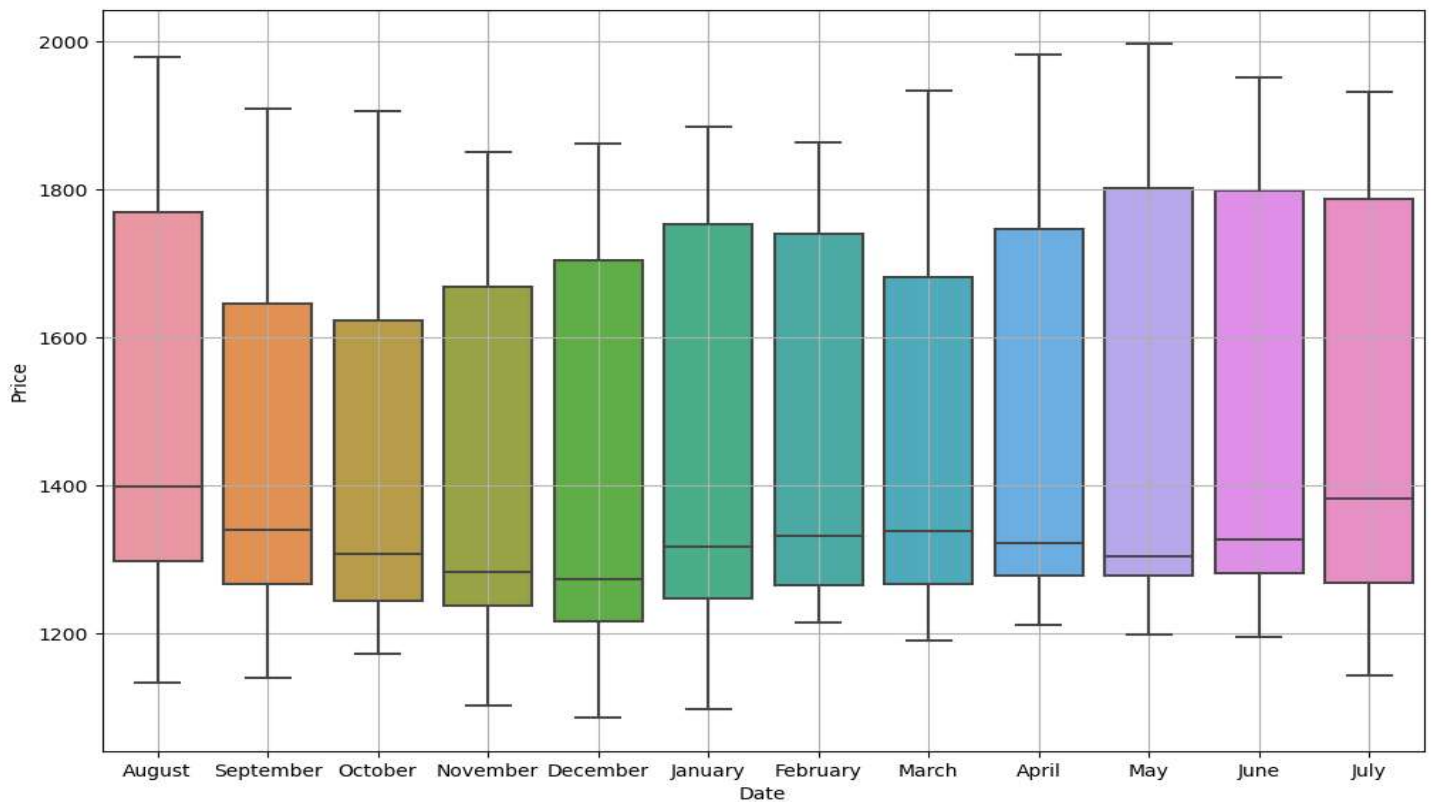


Figure 4

Date	April	August	December	February	January	July	June	March	May	November	October	September
Date												
2013	NaN	1398.908435	1235.049289	NaN	NaN	NaN	NaN	NaN	NaN	1273.433333	1324.064537	1348.46
2014	1306.722016	1296.680952	1211.151493	1311.909217	1242.034783	1318.708015	1289.651636	1342.099255	1296.035652	1191.374217	1233.464537	1247.03
2015	1211.949289	1133.542112	1086.686276	1237.649217	1261.790198	1143.381928	1194.826561	1190.208380	1198.476190	1102.213540	1172.844743	1139.39
2016	1254.056398	1344.482609	1153.690476	1215.149217	1097.289474	1344.494217	1279.190909	1256.022016	1272.646874	1249.937350	1275.527826	1330.68
2017	1271.136842	1298.121059	1280.014217	1248.641282	1209.089217	1238.990000	1272.422016	1241.055841	1255.344743	1292.513540	1290.017470	1325.51
2018	1338.661905	1217.851493	1266.209217	1332.594737	1340.518302	1251.389731	1291.846874	1333.737350	1313.590198	1234.037350	1229.434102	1201.99
2019	1297.146874	1511.818182	1484.347619	1330.272861	1294.500000	1420.322016	1363.440000	1301.871429	1294.403834	1471.104217	1497.634783	1511.44
2020	1703.266667	1980.271429	1862.845455	1593.441282	1562.585714	1831.667470	1743.809091	1596.859091	1710.889217	1851.149217	1905.550000	1908.92
2021	1760.909524	1774.994743	1776.981107	1788.609703	1867.557895	1807.623810	1816.403834	1709.221059	1851.620000	1807.032588	1777.428571	1763.39
2022	1937.545000	1765.860189	1807.657143	1839.630755	1817.750000	1725.284217	1838.780952	1933.273233	1832.684969	1734.533333	1665.061160	1690.05
2023	1983.525492	1905.429565	NaN	1864.415789	1884.424217	1931.594217	1951.438095	1895.538450	1997.940909	NaN	NaN	

Table 7- Pivot table to display monthly price across years

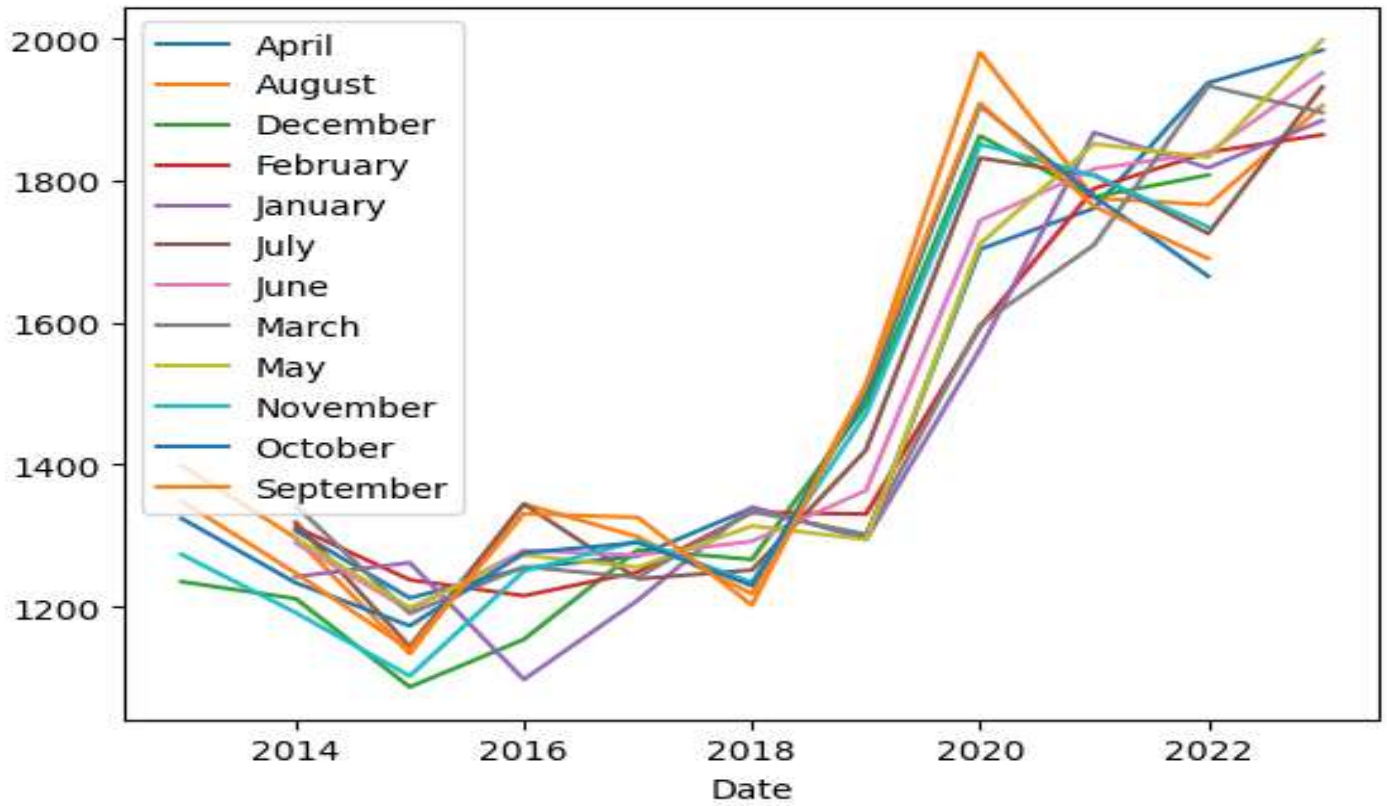


Figure 5

-Decomposition

#Decompose the time series additively

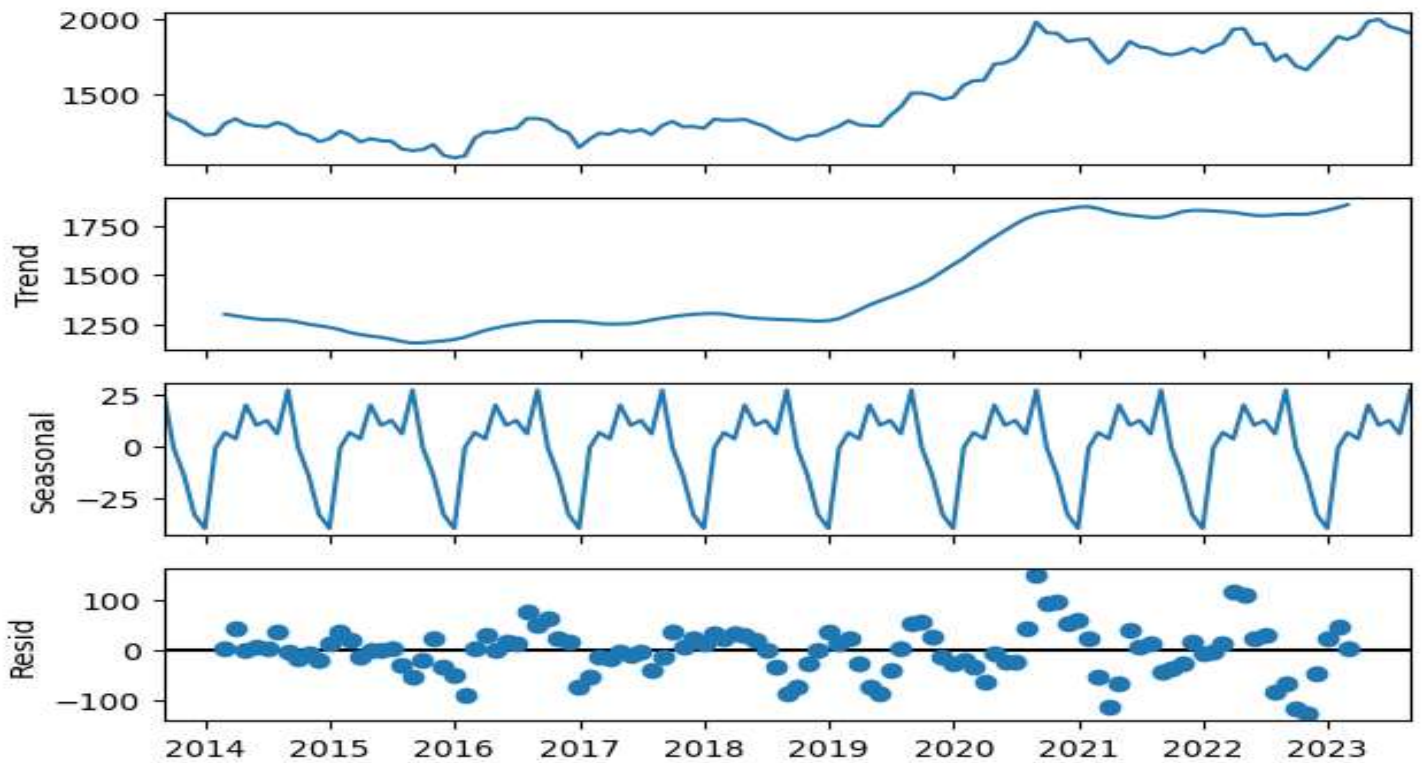


Figure 6

#Decompose the time series multiplicative

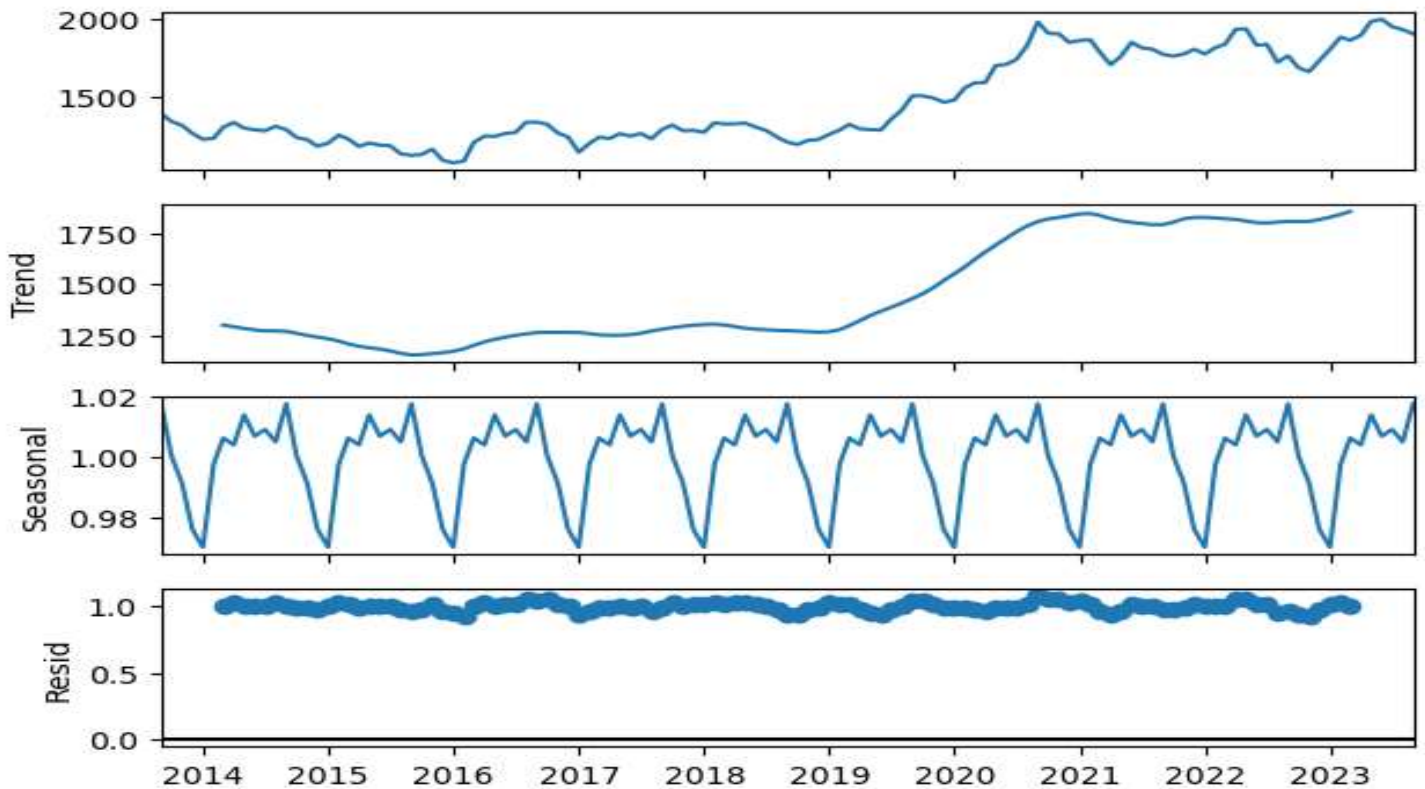


Figure 7

>Overall Observations:

- From the additively decomposition we see that the Trend is upward means that there are growth in the time series. And same as the Multiplicative
- The seasonal component shows the repeating cycle pattern of the time series.
- We see that there are random repeated noise shows in the time series.

2- Data Pre-processing:

-Split the data into train and test and plot the training and test data

First few rows of Training Data

Date	Price
2013-08-31	1398.908435
2013-09-30	1348.461905
2013-10-31	1324.064537
2013-11-30	1273.433333
2013-12-31	1235.049289

Table 8

Last few rows of Training Data

Date	Price
2020-08-31	1980.271429
2020-09-30	1908.923064
2020-10-31	1905.550000
2020-11-30	1851.149217
2020-12-31	1862.845455

Table 9

First few rows of Test Data

Date	Price
2021-01-31	1867.557895
2021-02-28	1788.609703
2021-03-31	1709.221059
2021-04-30	1760.909524
2021-05-31	1851.620000

Table 10

Last few rows of Test Data

Date	Price
2023-04-30	1983.525492
2023-05-31	1997.940909
2023-06-30	1951.438095
2023-07-31	1931.594217
2023-08-31	1905.429565

Table 11

>Observations:

- Train data has present 89 rows and 1 column.
- Test data has present 32 rows and 1 column.

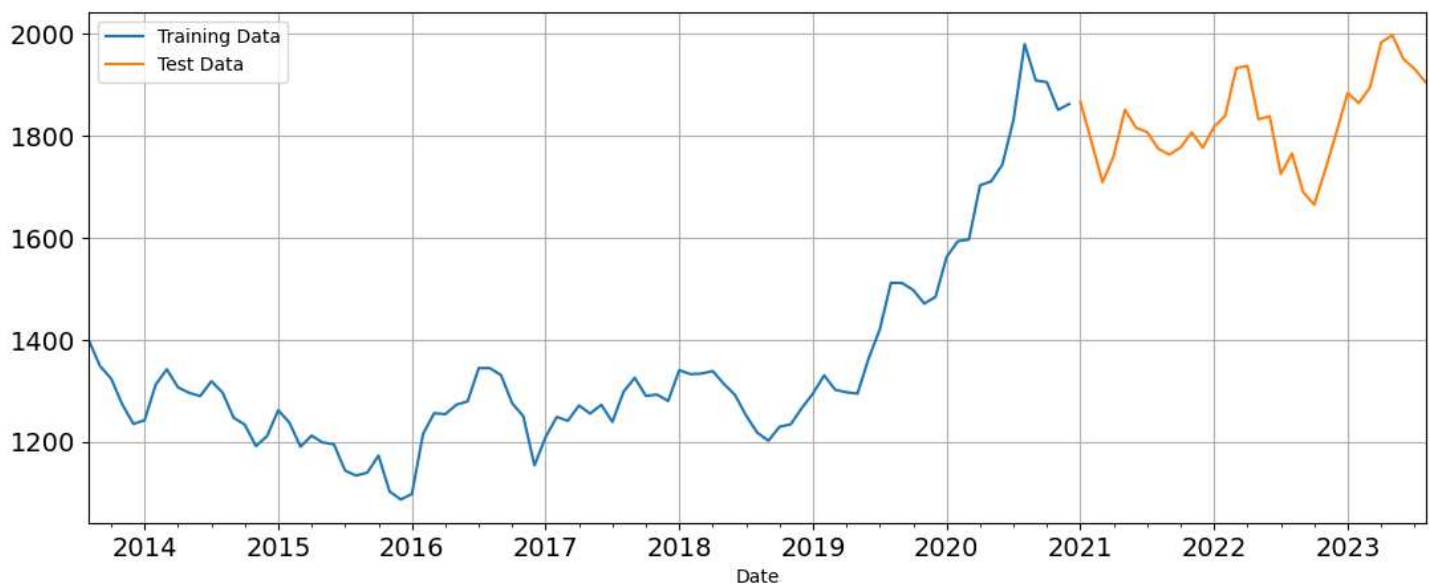


Figure 8

>Observations:

- From the above plot we see that the training data shows a general upward trend starting from around 2017 after a period of fluctuation and decline. The sharpest increase is observed between 2019 and 2020, indicating a significant rise in the data values during this period.
- From the above plot we see that the test data continues to show an overall upward trend, but with noticeable fluctuations. After reaching a peak around late 2022, the values start to decrease slightly but still remain higher than the early parts of the test period.

3- Model Building - Original Data:

#Numerical time instance order for both the training and test set

Training Time instance

[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89]

Test Time instance

[89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120]

First few rows of Training Data

	Price	time
Date		
2013-08-31	1398.908435	1
2013-09-30	1348.461905	2
2013-10-31	1324.064537	3
2013-11-30	1273.433333	4
2013-12-31	1235.049289	5

Last few rows of Training Data

	Price	time
Date		
2020-08-31	1980.271429	85
2020-09-30	1908.923064	86
2020-10-31	1905.550000	87
2020-11-30	1851.149217	88
2020-12-31	1862.845455	89

Table 12

First few rows of Test Data

	Price	time
Date		
2021-01-31	1867.557895	89
2021-02-28	1788.609703	90
2021-03-31	1709.221059	91
2021-04-30	1760.909524	92
2021-05-31	1851.620000	93

Last few rows of Test Data

	Price	time
Date		
2023-04-30	1983.525492	116
2023-05-31	1997.940909	117
2023-06-30	1951.438095	118
2023-07-31	1931.594217	119
2023-08-31	1905.429565	120

Table 13

-Linear Regression

Now that our training and test data has been modified, let us go ahead use *LinearRegression* to build the model on the training data and test the model on the test data.*

LinearRegression
LinearRegression()

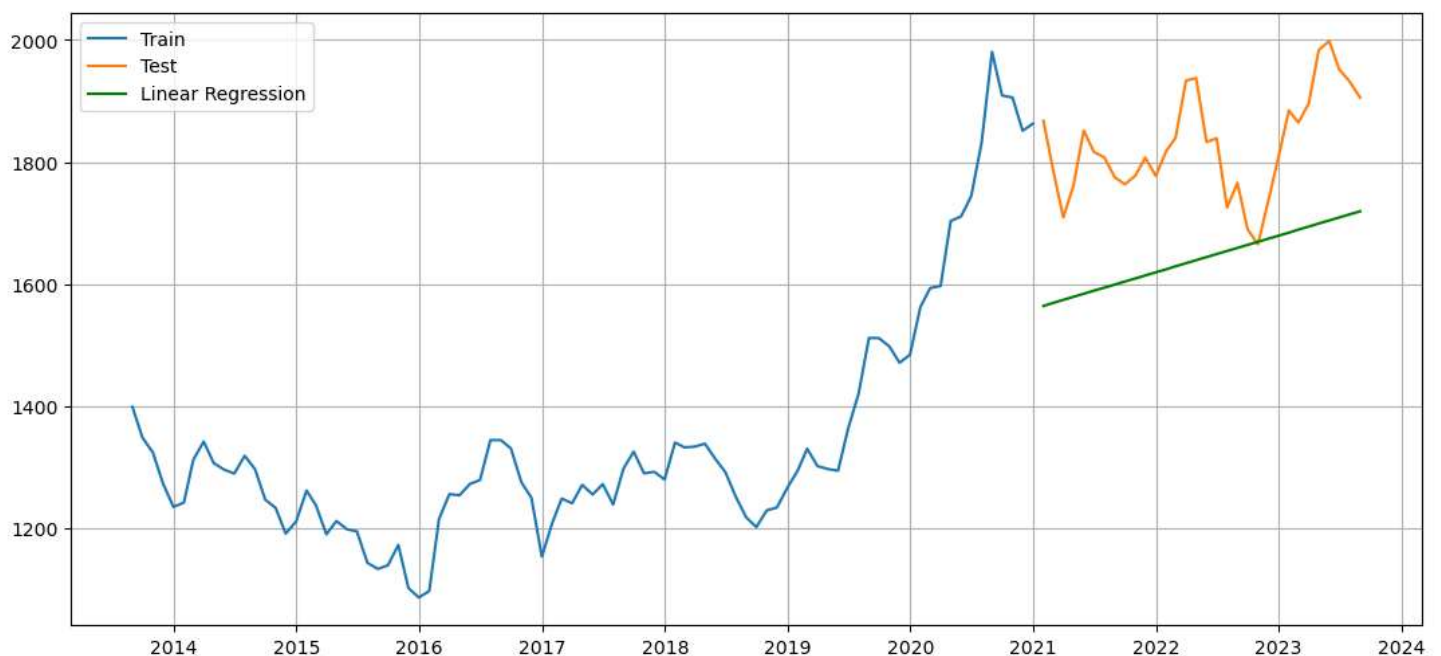


Figure 9

>Observations:

- For RegressionOnTime forecast on the Test Data, RMSE is 201.18
- From the above plot we say that the linear regression line which is green appears to be an oversimplified model for predicting future values, as it doesn't capture the significant fluctuations observed in the actual test data which is orange. The actual test values deviate substantially from the linear trend, indicating that a more complex model might be needed to account for the variability in the data.
- From the above plot we say that the training data which is blue shows a significant upward trend starting around 2020. Before that, the data fluctuates without a clear long-term trend. The test data continues to show volatility, with peaks and troughs, but the overall level seems higher than in the earlier part of the dataset.

-Moving Average (MA)

For the moving average model, we are going to calculate rolling means (or moving averages) for different intervals. The best interval can be determined by the maximum accuracy (or the minimum error) over here. For Moving Average, we are going to average over the entire data.

Price	
Date	
2013-08-31	1398.908435
2013-09-30	1348.461905
2013-10-31	1324.064537
2013-11-30	1273.433333
2013-12-31	1235.049289

Price		Trailing_2	Trailing_4	Trailing_6	Trailing_9
Date					
2013-08-31	1398.908435	NaN	NaN	NaN	NaN
2013-09-30	1348.461905	1373.685170	NaN	NaN	NaN
2013-10-31	1324.064537	1336.263221	NaN	NaN	NaN
2013-11-30	1273.433333	1298.748935	1336.217052	NaN	NaN
2013-12-31	1235.049289	1254.241311	1295.252266	NaN	NaN

Table 14- Head of the movingaverage data

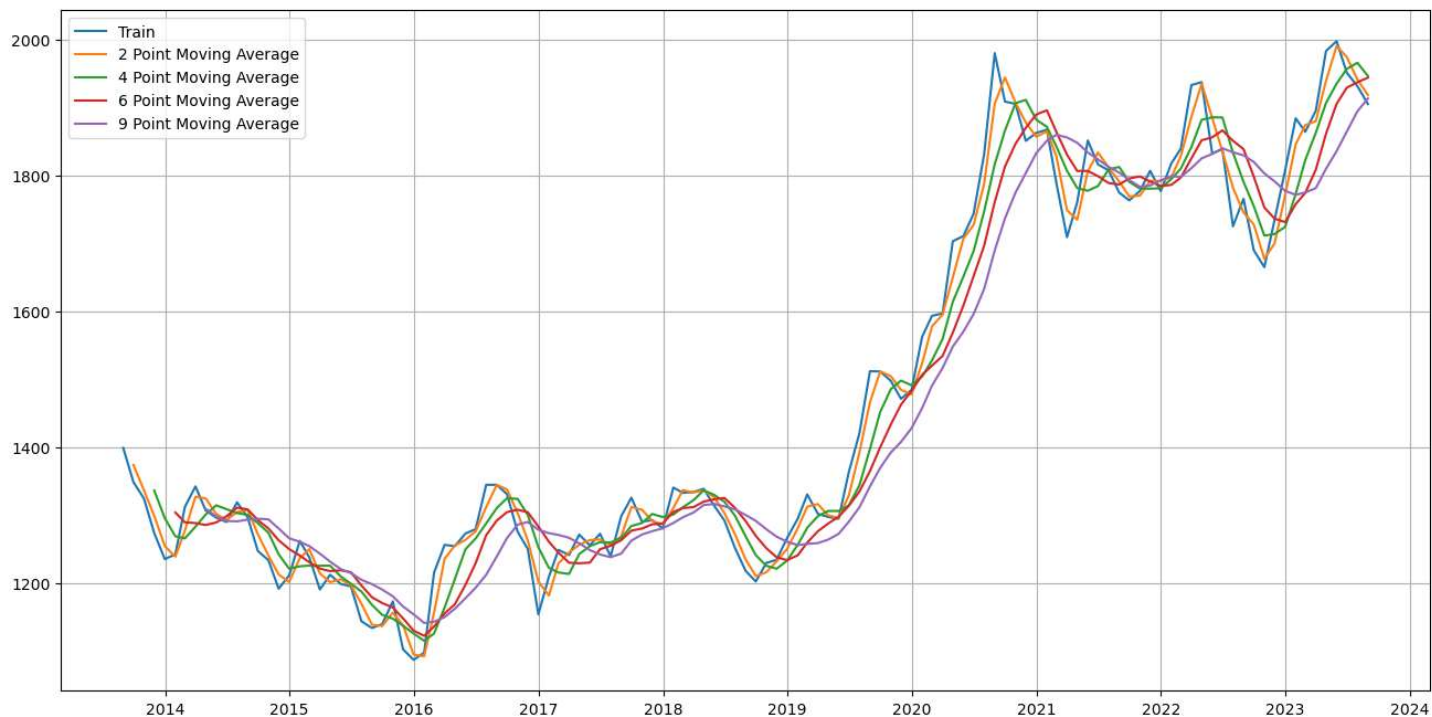


Figure 10

Lets split the data into train and test and plot this Time Series. The window of the moving average is need to be carefully selected as too big a window will result in not having any test set as the whole series might get averaged over.

For 2 point Moving Average Model forecast on the Testing Data, RMSE is 27.872
 For 4 point Moving Average Model forecast on the Testing Data, RMSE is 54.239
 For 6 point Moving Average Model forecast on the Testing Data, RMSE is 69.513
 For 9 point Moving Average Model forecast on the Testing Data, RMSE is 83.336

Test RMSE

Linear Regression	201.175903
2pointTrailingMovingAverage	27.871919
4pointTrailingMovingAverage	54.239048
6pointTrailingMovingAverage	69.512886
9pointTrailingMovingAverage	83.336308

Before we go on to build the various Exponential Smoothing models, let us plot all the best models and compare the Time Series plots.



Figure 11

>Observations:

- We observe from the above plot that the best moving average model seems to closely follow the test data, indicating it captures the short-term fluctuations in the test data quite well. In contrast, the linear regression model (green line) shows a much simpler, upward-sloping trend line, which fails to capture the short-term variations observed in the test data.
- From the plot we see that the test data continues this upward trend but with more pronounced oscillations. This indicates that while the overall trend is upward, the model faces challenges in predicting the short-term volatility that becomes more evident in the test period.
- Overall the moving average model may be more suitable for capturing short-term patterns, while linear regression is better suited for capturing long-term trends.

-Simple Exponential Smoothing Models

```
{'smoothing_level': 0.995,
'smoothing_trend': nan,
'smoothing_seasonal': nan,
'damping_trend': nan,
'initial_level': 1398.9084349593495,
'initial_trend': nan,
'initial_seasons': array([], dtype=float64),
'use_boxcox': False,
'lamda': None,
'remove_bias': False}
```

Date	Price	predict
2021-01-31	1867.557895	1862.788334
2021-02-28	1788.609703	1862.788334
2021-03-31	1709.221059	1862.788334
2021-04-30	1760.909524	1862.788334
2021-05-31	1851.620000	1862.788334

Table 15- Head of simple exponential models

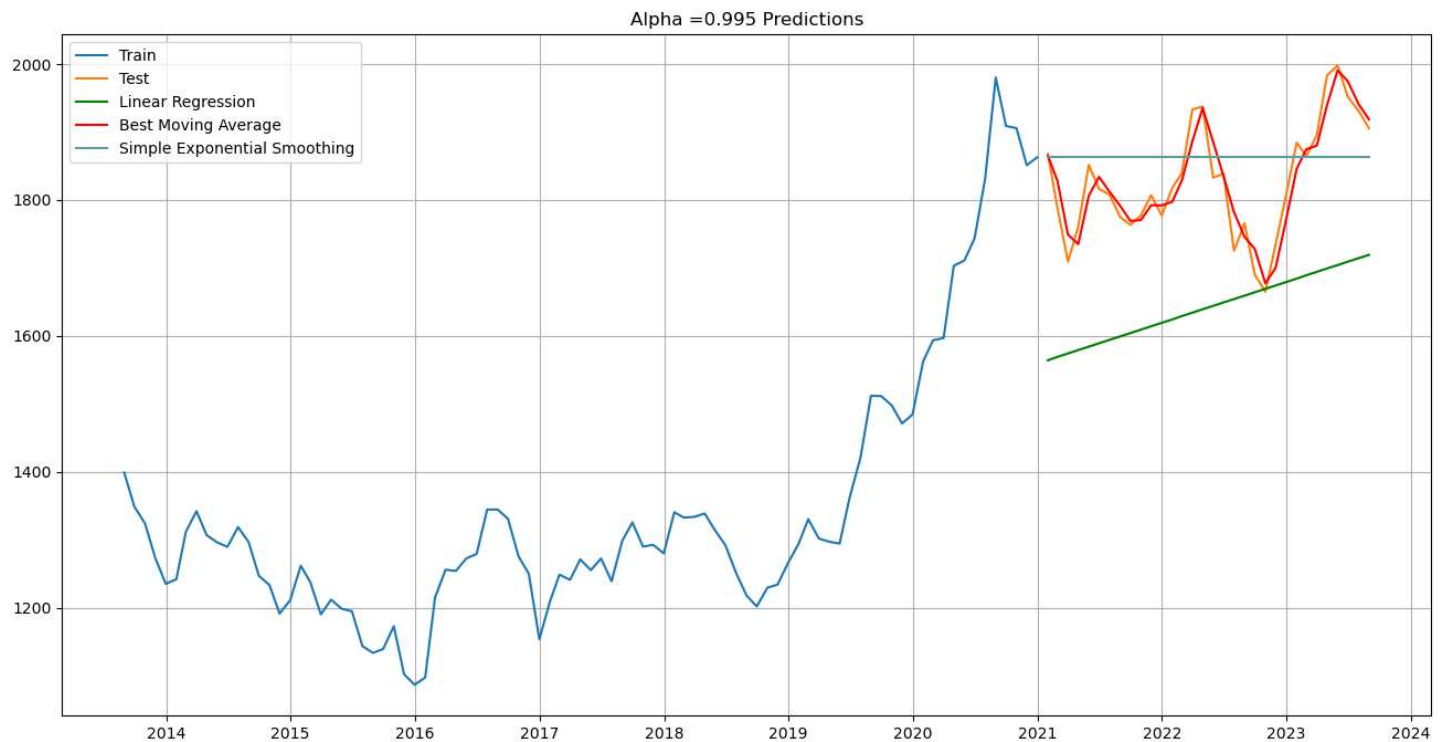


Figure 12

For Alpha =0.995 Simple Exponential Smoothing Model forecast on the Test Data, RMSE is 90.187

Test RMSE

Linear Regression	201.175903
2pointTrailingMovingAverage	27.871919
4pointTrailingMovingAverage	54.239048
6pointTrailingMovingAverage	69.512886
9pointTrailingMovingAverage	83.336308
Alpha=0.995,SimpleExponentialSmoothing	90.186647

>Observations:

-Double Exponential Smoothing (Holt's Model)

Two parameters α and β are estimated in this model. Level and Trend are accounted for in this model.

	Alpha Values	Beta Values	Train RMSE	Test RMSE
61	1.0	0.8	51.956119	91.155144
54	0.9	0.9	51.786497	94.424902
40	0.8	0.3	47.464513	98.549464
57	1.0	0.4	46.863742	99.605155
48	0.9	0.3	46.391811	103.510659

Table 16- Head of Double exponential model



Figure 13

	Test RMSE
Linear Regression	201.175903
2pointTrailingMovingAverage	27.871919
4pointTrailingMovingAverage	54.239048
6pointTrailingMovingAverage	69.512886
9pointTrailingMovingAverage	83.336308
Alpha=0.995,SimpleExponentialSmoothing	90.186647
Alpha=0.9,Beta=0.3,DoubleExponentialSmoothing	91.155144

-Triple Exponential Smoothing (Holt - Winter's Model)

Three parameters α , β and γ are estimated in this model. Level, Trend and Seasonality are accounted for in this model.

```
{'smoothing_level': 0.9999999850988388,  
'smoothing_trend': 0.06715884181152897,  
'smoothing_seasonal': 1.0091872842263913e-10,  
'damping_trend': nan,  
'initial_level': 1356.382364654156,  
'initial_trend': -5.636954782955098,  
'initial_seasons': array([ 48.10828597,  20.91200841,   3.84520197, -35.76444479,  
-53.51949843, -17.15488926,  14.75283201,   7.84355426,  
 19.00414747,   6.56431213,  13.15030297,  22.22622574]),  
'use_boxcox': False,  
'lamda': None,  
'remove_bias': False}
```

Price auto_predict

Date

2021-01-31	1867.557895	1921.810997
2021-02-28	1788.609703	1976.319651
2021-03-31	1709.221059	1992.011307
2021-04-30	1760.909524	2025.772833
2021-05-31	1851.620000	2035.933931



Figure 14

	Alpha Values	Beta Values	Gamma Values	Train RMSE	Test RMSE
355	0.8	0.7	0.6	71.777638	94.104803
297	0.7	0.8	0.4	57.325899	100.230008
404	0.9	0.5	0.7	79.889471	102.354309
338	0.8	0.5	0.5	55.269756	104.018940
248	0.6	1.0	0.3	56.528514	104.240545

Table 17- Head of the Triple exponential model



Figure 15

	Test RMSE
Linear Regression	201.175903
2pointTrailingMovingAverage	27.871919
4pointTrailingMovingAverage	54.239048
6pointTrailingMovingAverage	69.512886
9pointTrailingMovingAverage	83.336308
Alpha=0.995, SimpleExponentialSmoothing	90.186647
Alpha=0.9, Beta=0.3, DoubleExponentialSmoothing	91.155144
Alpha=0.676, Beta=0.088, Gamma=0.323, TripleExponentialSmoothing	504.694477
Alpha=0.8, Beta=0.5, Gamma=0.5, TripleExponentialSmoothing	94.104803

	Test RMSE
2pointTrailingMovingAverage	27.871919
4pointTrailingMovingAverage	54.239048
6pointTrailingMovingAverage	69.512886
9pointTrailingMovingAverage	83.336308
Alpha=0.995, SimpleExponentialSmoothing	90.186647
Alpha=0.9, Beta=0.3, DoubleExponentialSmoothing	91.155144
Alpha=0.8, Beta=0.5, Gamma=0.5, TripleExponentialSmoothing	94.104803
Linear Regression	201.175903
Alpha=0.676, Beta=0.088, Gamma=0.323, TripleExponentialSmoothing	504.694477

4- Check for Stationarity:

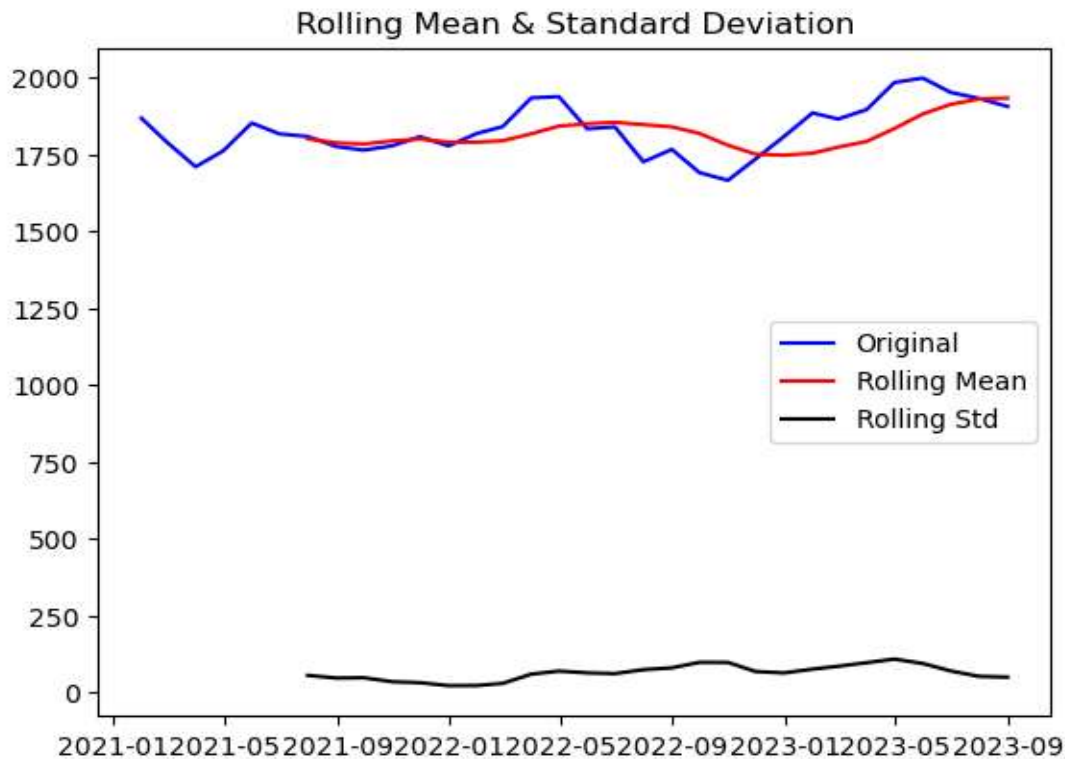


Figure 16

```
Results of Dickey-Fuller Test:
Test Statistic      -3.237430
p-value             0.017918
#Lags Used          5.000000
Number of Observations Used 26.000000
Critical Value (1%) -3.711212
Critical Value (5%) -2.981247
Critical Value (10%) -2.630095
dtype: float64
```

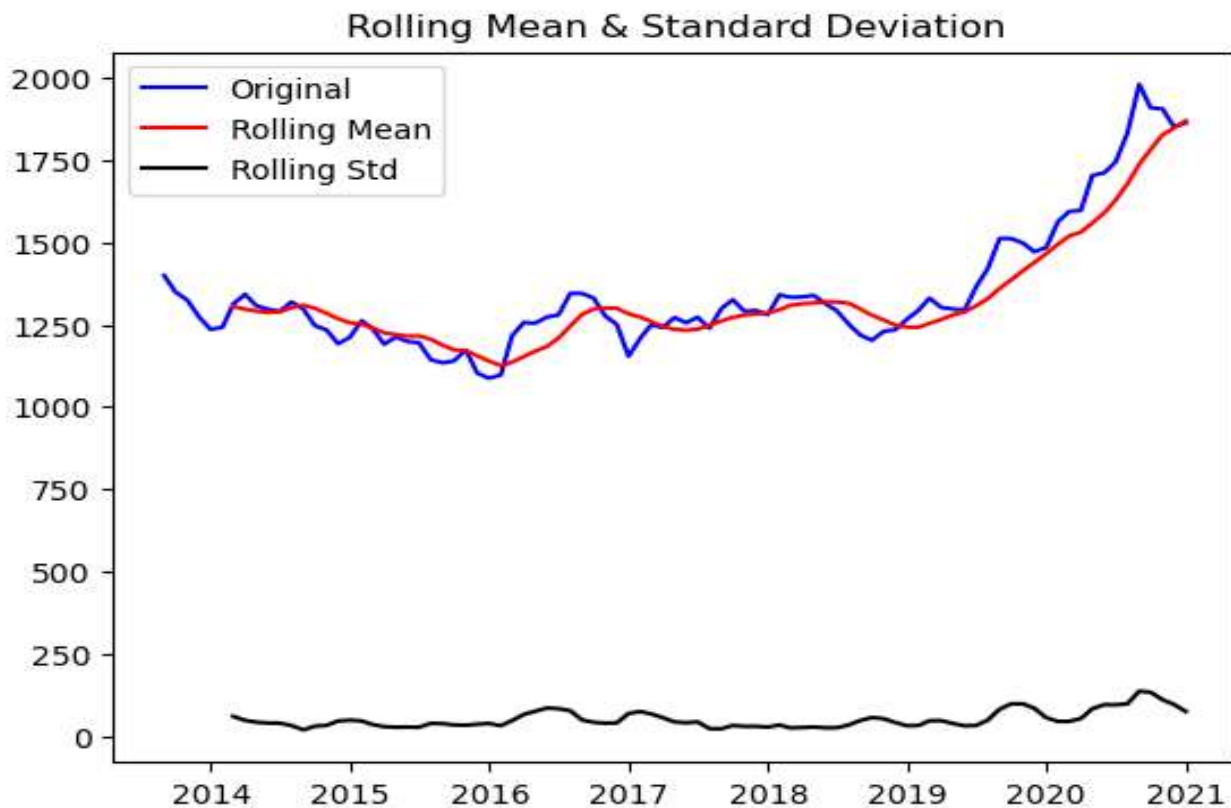


Figure 17

Results of Dickey-Fuller Test:

Test Statistic	0.662224
p-value	0.989053
#Lags Used	0.000000
Number of Observations Used	88.000000
Critical Value (1%)	-3.506944
Critical Value (5%)	-2.894990
Critical Value (10%)	-2.584615

dtype: float64

5- Model Building - Stationary Data:

Plot the Autocorrelation and the Partial Autocorrelation function plots on the whole data.

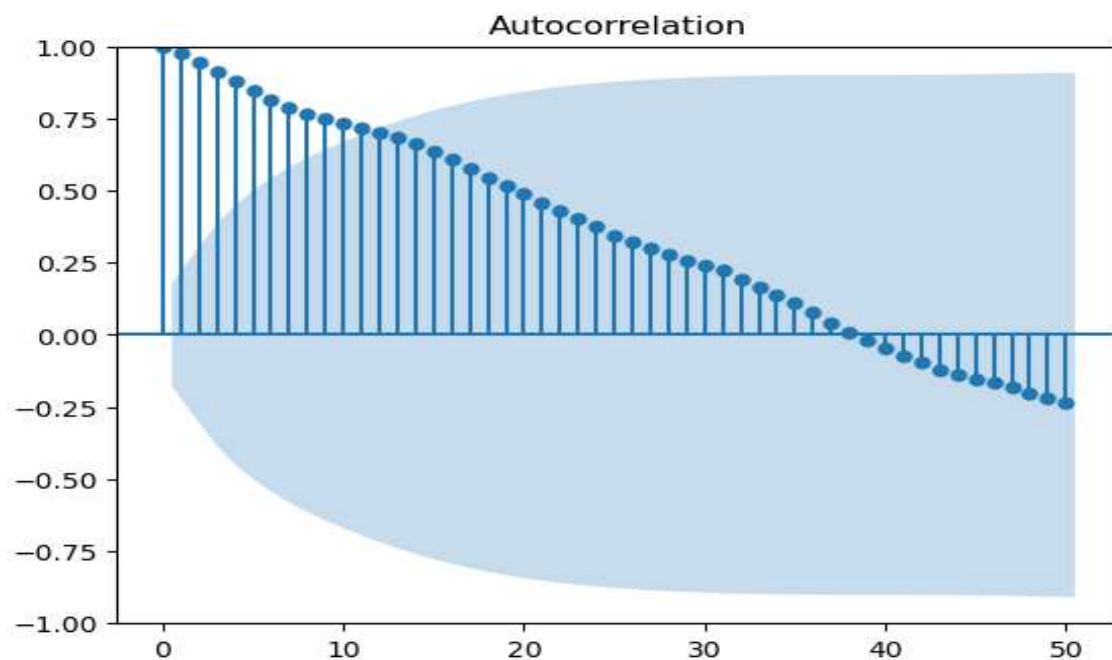


Figure 18

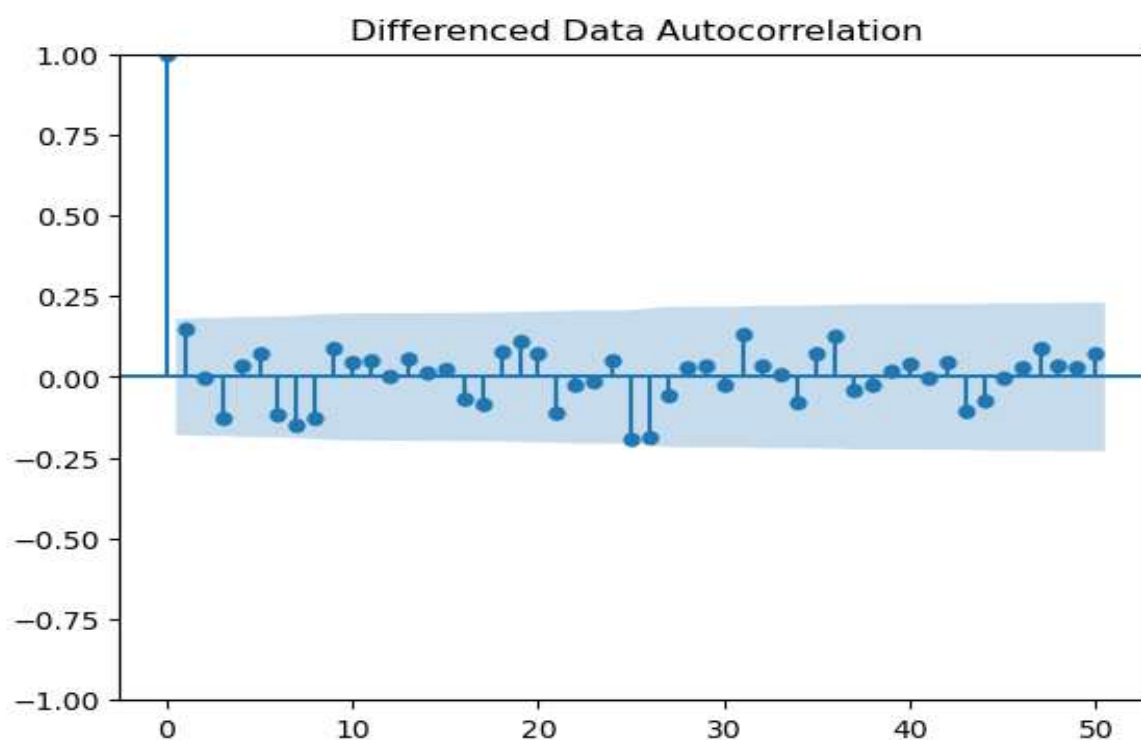


Figure 19

-Auto ARIMA Model

Some parameter combinations for the Model...

Model: (0, 1, 1)

Model: (0, 1, 2)

Model: (1, 1, 0)

Model: (1, 1, 1)

Model: (1, 1, 2)

Model: (2, 1, 0)

Model: (2, 1, 1)

Model: (2, 1, 2)

#AIC values of Auto_ARIMA model

```
ARIMA(0, 1, 0) - AIC:917.4293185297809
ARIMA(0, 1, 1) - AIC:916.4722743708096
ARIMA(0, 1, 2) - AIC:917.6698876683952
ARIMA(1, 1, 0) - AIC:916.1677536171871
ARIMA(1, 1, 1) - AIC:918.1651665718056
ARIMA(1, 1, 2) - AIC:917.4792443965465
ARIMA(2, 1, 0) - AIC:918.1615701949067
ARIMA(2, 1, 1) - AIC:918.818780687494
ARIMA(2, 1, 2) - AIC:918.4918477196446
```

Sort the above AIC values in the ascending order to get the parameters for the minimum AIC value

	param	AIC
3	(1, 1, 0)	916.167754
1	(0, 1, 1)	916.472274
0	(0, 1, 0)	917.429319
5	(1, 1, 2)	917.479244
2	(0, 1, 2)	917.669888
6	(2, 1, 0)	918.161570
4	(1, 1, 1)	918.165167
8	(2, 1, 2)	918.491848
7	(2, 1, 1)	918.818781

SARIMAX Results

```
=====
Dep. Variable:          Price      No. Observations:          89
Model:                ARIMA(1, 1, 0)  Log Likelihood          -456.084
Date:                Sun, 11 Aug 2024  AIC                      916.168
Time:                14:34:58         BIC                      921.122
Sample:              08-31-2013       HQIC                     918.164
                        - 12-31-2020
Covariance Type:      opg
=====
```

	coef	std err	z	P> z	[0.025	0.975]
ar.L1	0.1911	0.083	2.294	0.022	0.028	0.354
sigma2	1857.4559	242.134	7.671	0.000	1382.882	2332.030

```
=====
Ljung-Box (L1) (Q):          0.01  Jarque-Bera (JB):          4.39
Prob(Q):                    0.91  Prob(JB):              0.11
Heteroskedasticity (H):      2.17  Skew:                  0.47
Prob(H) (two-sided):         0.04  Kurtosis:              3.56
=====
```

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

Predict on the Test Set using this model and evaluate the model.

2021-01-31	1865.081152
2021-02-28	1865.508498
2021-03-31	1865.590184
2021-04-30	1865.605798
2021-05-31	1865.608782
2021-06-30	1865.609353
2021-07-31	1865.609462
2021-08-31	1865.609483
2021-09-30	1865.609487
2021-10-31	1865.609487
2021-11-30	1865.609487
2021-12-31	1865.609488
2022-01-31	1865.609488
2022-02-28	1865.609488
2022-03-31	1865.609488
2022-04-30	1865.609488
2022-05-31	1865.609488
2022-06-30	1865.609488
2022-07-31	1865.609488
2022-08-31	1865.609488
2022-09-30	1865.609488
2022-10-31	1865.609488
2022-11-30	1865.609488
2022-12-31	1865.609488
2023-01-31	1865.609488
2023-02-28	1865.609488
2023-03-31	1865.609488
2023-04-30	1865.609488
2023-05-31	1865.609488
2023-06-30	1865.609488
2023-07-31	1865.609488
2023-08-31	1865.609488

Freq: M, Name: predicted_mean, dtype: float64



Figure 20

RMSE on the Auto_ARIMA model is 91.300619.

-ARIMA Model

Some parameter combinations for the Model...

Model: (0, 0, 1)
Model: (0, 0, 2)
Model: (0, 1, 0)
Model: (0, 1, 1)
Model: (0, 1, 2)
Model: (0, 2, 0)
Model: (0, 2, 1)
Model: (0, 2, 2)
Model: (1, 0, 0)
Model: (1, 0, 1)
Model: (1, 0, 2)
Model: (1, 1, 0)
Model: (1, 1, 1)
Model: (1, 1, 2)
Model: (1, 2, 0)
Model: (1, 2, 1)
Model: (1, 2, 2)
Model: (2, 0, 0)
Model: (2, 0, 1)
Model: (2, 0, 2)
Model: (2, 1, 0)
Model: (2, 1, 1)
Model: (2, 1, 2)
Model: (2, 2, 0)
Model: (2, 2, 1)
Model: (2, 2, 2)

#AIC values of ARIMA model

ARIMA(0, 0, 0) - AIC:1193.7374259326643
ARIMA(0, 0, 1) - AIC:1102.959679803812
ARIMA(0, 0, 2) - AIC:1034.0375805574831
ARIMA(0, 1, 0) - AIC:917.4293185297809
ARIMA(0, 1, 1) - AIC:916.4722743708096
ARIMA(0, 1, 2) - AIC:917.6698876683952
ARIMA(0, 2, 0) - AIC:949.0223047751825
ARIMA(0, 2, 1) - AIC:910.6364261937139
ARIMA(0, 2, 2) - AIC:910.7788749317984
ARIMA(1, 0, 0) - AIC:934.7439572466803
ARIMA(1, 0, 1) - AIC:933.5521487151823
ARIMA(1, 0, 2) - AIC:934.4648736669429
ARIMA(1, 1, 0) - AIC:916.1677536171871
ARIMA(1, 1, 1) - AIC:918.1651665718056
ARIMA(1, 1, 2) - AIC:917.4792443965465
ARIMA(1, 2, 0) - AIC:935.1923267932311
ARIMA(1, 2, 1) - AIC:910.7416052565202
ARIMA(1, 2, 2) - AIC:912.7197555796479
ARIMA(2, 0, 0) - AIC:933.1134887969071
ARIMA(2, 0, 1) - AIC:935.0886145430908
ARIMA(2, 0, 2) - AIC:934.8600529519323
ARIMA(2, 1, 0) - AIC:918.1615701949067
ARIMA(2, 1, 1) - AIC:918.818780687494
ARIMA(2, 1, 2) - AIC:918.4918477196446
ARIMA(2, 2, 0) - AIC:934.0415190805284
ARIMA(2, 2, 1) - AIC:912.6616399054858
ARIMA(2, 2, 2) - AIC:913.6420585188939

Sort the above AIC values in the ascending order to get the parameters for the minimum AIC value

	param	AIC
16	(0, 2, 1)	910.636426
25	(1, 2, 1)	910.741605
17	(0, 2, 2)	910.778875
34	(2, 2, 1)	912.661640
26	(1, 2, 2)	912.719756
35	(2, 2, 2)	913.642059
21	(1, 1, 0)	916.167754
3	(1, 1, 0)	916.167754
1	(0, 1, 1)	916.472274
13	(0, 1, 1)	916.472274

SARIMAX Results

```
=====
Dep. Variable:          Price      No. Observations:          89
Model:                ARIMA(0, 2, 1)  Log Likelihood          -453.318
Date:                Sun, 11 Aug 2024  AIC                      910.636
Time:                14:35:40         BIC                      915.568
Sample:              08-31-2013       HQIC                     912.622
                  - 12-31-2020

Covariance Type:          opg
=====
              coef      std err          z      P>|z|      [0.025      0.975]
-----
ma.L1          -0.9420        0.031     -30.783      0.000       -1.002       -0.882
sigma2         1918.9392     255.009       7.525      0.000     1419.131     2418.748
=====
Ljung-Box (L1) (Q):                1.18   Jarque-Bera (JB):                2.40
Prob(Q):                          0.28   Prob(JB):                  0.30
Heteroskedasticity (H):            1.48   Skew:                      0.33
Prob(H) (two-sided):              0.30   Kurtosis:                  3.48
=====
```

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

#Forecasted the ARIMA model

2021-01-31	1880.626516
2021-02-28	1898.407577
2021-03-31	1916.188638
2021-04-30	1933.969699
2021-05-31	1951.750760
2021-06-30	1969.531821
2021-07-31	1987.312881
2021-08-31	2005.093942
2021-09-30	2022.875003
2021-10-31	2040.656064
2021-11-30	2058.437125
2021-12-31	2076.218186
2022-01-31	2093.999247
2022-02-28	2111.780308
2022-03-31	2129.561369
2022-04-30	2147.342430
2022-05-31	2165.123491
2022-06-30	2182.904552
2022-07-31	2200.685613
2022-08-31	2218.466674
2022-09-30	2236.247735
2022-10-31	2254.028796
2022-11-30	2271.809857
2022-12-31	2289.590918
2023-01-31	2307.371979
2023-02-28	2325.153040
2023-03-31	2342.934101
2023-04-30	2360.715162
2023-05-31	2378.496223
2023-06-30	2396.277284
2023-07-31	2414.058345
2023-08-31	2431.839406

Freq: M, Name: predicted_mean, dtype: float64



Figure 21

RMSE	
Auto_ARIMA	91.300619
ARIMA_model1	359.605554

Test RMSE	
Linear Regression	201.175903
2pointTrailingMovingAverage	27.871919
4pointTrailingMovingAverage	54.239048
6pointTrailingMovingAverage	69.512886
9pointTrailingMovingAverage	83.336308
Alpha=0.995, SimpleExponential Smoothing	90.186647
Alpha=0.9, Beta=0.3, DoubleExponential Smoothing	91.155144
Alpha=0.676, Beta=0.088, Gamma=0.323, TripleExponential Smoothing	504.694477
Alpha=0.8, Beta=0.5, Gamma=0.5, TripleExponential Smoothing	94.104803

6- Compare the performance of the models:

RMSE	
Linear Regression	201.175903
2pointTrailingMovingAverage	27.871919
4pointTrailingMovingAverage	54.239048
6pointTrailingMovingAverage	69.512886
9pointTrailingMovingAverage	83.336308
Alpha=0.995, SimpleExponential Smoothing	90.186647
Alpha=0.9, Beta=0.3, DoubleExponential Smoothing	91.155144
Alpha=0.676, Beta=0.088, Gamma=0.323, TripleExponential Smoothing	504.694477
Alpha=0.8, Beta=0.5, Gamma=0.5, TripleExponential Smoothing	94.104803
Auto_ARIMA	91.300619
ARIMA_model1	359.605554

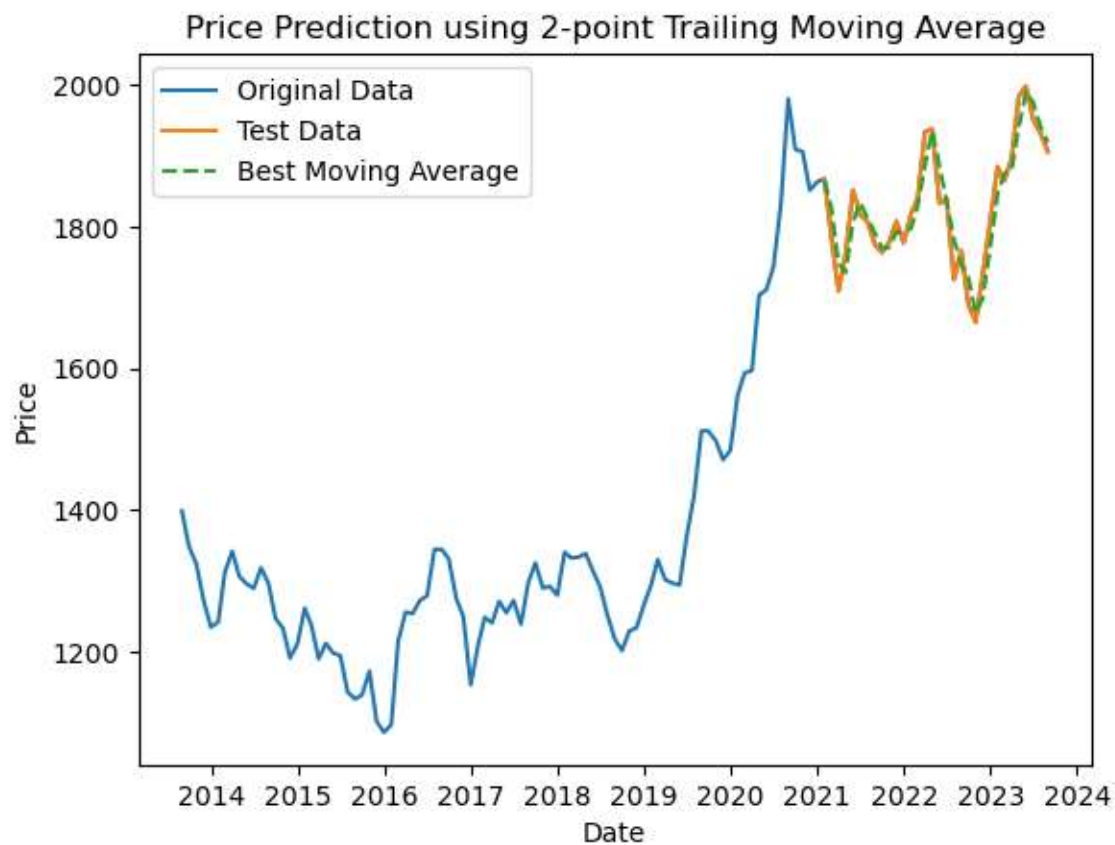


Figure 22

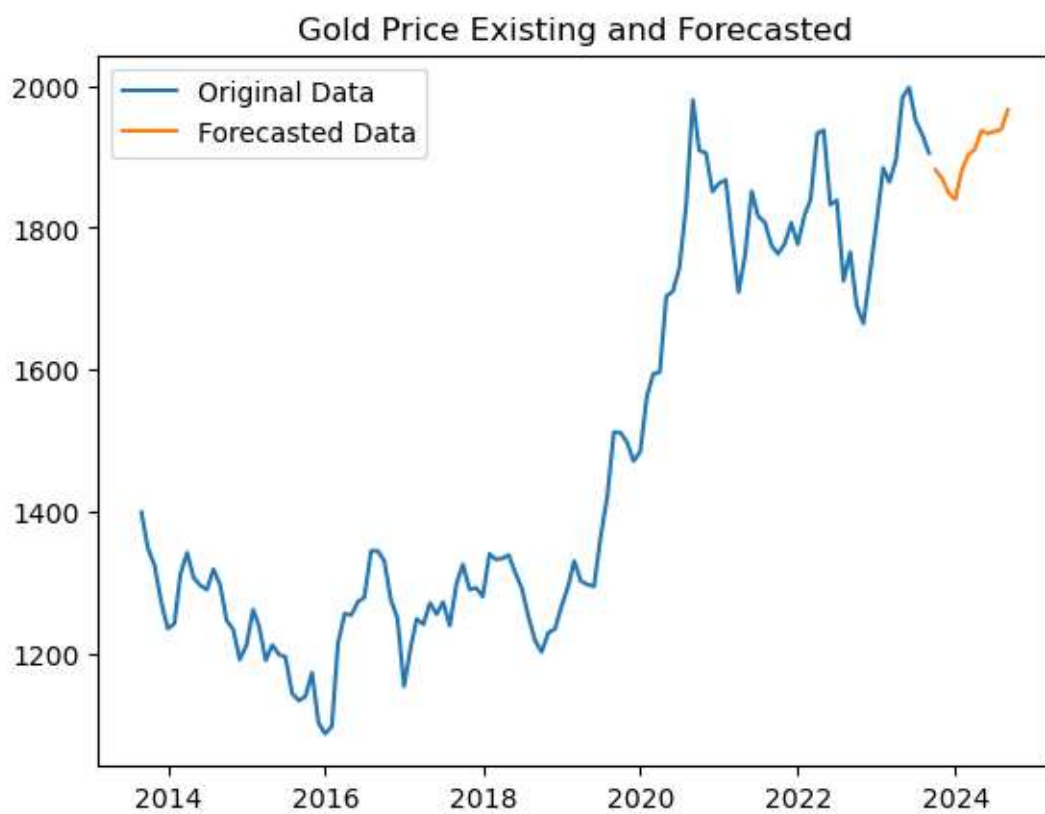


Figure 23

7- Actionable Insights & Recommendations:

-Insights

From all above analysis we come out with some insights which are given below;

- The use of the ExponentialSmoothing model with both an additive trend and seasonal component suggests that the gold prices have both a consistent upward or downward trend and a repeating seasonal pattern. The fitted model will capture these components, providing a more accurate forecast by accounting for historical trends and seasonality.
- The missing value in the dataset could potentially affect the model's performance if not handled correctly. If this missing value is not imputed or managed, it might lead to biased or inaccurate forecasts, as the model relies on complete data to understand the trend and seasonality.

-Recommendations

From all above analysis we come out with some recommendations which are given below;

- Before fitting the model, address the missing value in the dataset. You could impute the missing value using interpolation, forward fill, or backward fill methods to ensure the model has a complete dataset to work with. This will improve the model's accuracy and reliability.
- After fitting the model and generating the forecast, evaluate its performance by comparing the forecasted values to a validation set or by using statistical measures like Mean Absolute Error or Root Mean Squared Error. This evaluation will help in assessing whether the model is accurately capturing the underlying trend and seasonality or if further tuning is required.